Network Adapter:

A network adapter is the component of a computer's internal hardware that is used for communicating over a network with another computer. It enable a computer to connect with another computer, server or any networking device over an LAN connection. A network adapter can be used over a wired or wireless network.

Network Router:

Resides at the Network Layer. Data Transmission Form is a Packet. Uses IP address. Used for connecting two or more networks. A router is a networking device that connects a local network to other local networks. Routers connect two or more logical subnets, which do not necessarily map one-to-one to the physical interfaces of the router.

Network Bridge:

A network bridge is a computer networking device that creates a single aggregate network from multiple communication networks or network segments. This function is called network bridging. Bridging is distinct from routing. Routing allows multiple networks to communicate independently and yet remain separate, whereas bridging connects two separate networks as if they were a single network. In the OSI model, bridging is performed in the data link layer (layer 2).

Network Switch:

Resides at the Data Link Layer. Data transmission form is a Frame. Uses MAC address. Used for connecting two or more nodes in the same network (L2) or different network (L3). A network switch is a computer networking device that is used to connect many devices together on a computer network. A switch is considered more advanced than a hub because a switch will send on a msg to device that needs or request it. Allow connections to multiple devices, manage ports, manage VLAN security settings.

Network Hub (or Repeater):

Also called repeaters — are even less advanced that switches; while a hub broadcasts the same data to all its ports, a network switch forwards data only to those devices that the data is intended for. Network hubs do not manage any traffic coming through them; they only broadcast — or repeat — packets from an incoming port to all other ports.